

Data-driven modeling of potato tuber size distributions

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The growth and development of a potato crop is a complex dynamical process, affected by a broad range of environmental and physiological factors. During the growing season, new tubers are formed and grow through different size classes. The resulting size distribution is an important factor for the market value of the crop and the industry would benefit from better knowledge about the factors the size distribution, but its development is difficult to predict due to the many interacting factors involved.

In this project, we will investigate the development of the tuber size distribution throughout the growing season as a function of environmental conditions. CropConsult, a company that provides advice to potato farmers, has collected field data containing tuber size distributions, crop measurements, weather data, soil information, and fertilization records. The goal is to use these data to identify which factors drive tuber formation and growth and to develop an interpretable dynamical model of this process.

A natural mathematical framework to model crop growth is a dynamical system

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t), \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^n$ describes the state of the crop and $\mathbf{u} \in \mathbb{R}^m$ contains environmental inputs such as temperature, water availability, and nutrient status. The governing function $\mathbf{f}$ is unknown and needs to be identified from data. To approximate $\mathbf{f}$, we use Sparse Identification of Nonlinear Dynamical Systems (SINDy), in which the dynamics are represented by a sparse combination of candidate model terms,

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \approx \mathbf{\Xi}^\top \boldsymbol{\theta}(\mathbf{x}, \mathbf{u}, t), \quad (2)$$

where $\boldsymbol{\theta} \in \mathbb{R}^p$ is a vector of candidate model terms and $\mathbf{\Xi} \in \mathbb{R}^{p \times n}$ is a sparse coefficient matrix to be identified from data. In order to introduce variability between tubers and plants, the SINDy coefficients $\mathbf{\Xi}$ will be treated as random parameters, $\mathbf{\Xi} \sim P_{\mathbf{\Xi}}$. Different realizations of these parameters correspond to different crop trajectories and thereby induce a distribution over the crop state.

The student will investigate how this probabilistic SINDy formulation can be used to describe the time evolution of the tuber size distribution. In particular, the student will study statistical moments such as the mean tuber size, its variance, and potentially higher-order moments. Starting from the SINDy representation, the student will derive or approximate dynamical equations for these moments and investigate suitable closure methods. Finally, the resulting models will be calibrated using field data and evaluated in their ability to reproduce the observed evolution of the tuber size distribution.

Reference for colloquium

J. Li and J. Chen, “Probability density evolution method: Background, significance and recent developments,” *Probabilistic Engineering Mechanics*, 44, 111–117, 2016.

Other references

S. L. Brunton, J. L. Proctor, and J. N. Kutz, “Discovering governing equations from data by sparse identification of nonlinear dynamical systems,” *Proceedings of the National Academy of Sciences*, 113(15), 3932–3937, 2016.